

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: MLA0303
COURSE CODE: Reinforcement learning

COURSE OUTCOME COVERED

CO1: Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems. (BL2, BL3)

Assessment Tool Weightage: 50%

Dr Dumpala Prasanth

Code of Conduct:

D. Hashini / 192425147 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



QUESTIONS: 10

Total Marks:50

Annexure A

Constraint - Based Problem-Solving

Duration: 2Hrs

1. Develop a Python-based Reinforcement Learning model using the OpenAI Gym environment to solve a Grid World navigation problem where the agent must reach the goal while avoiding obstacles. Explain how the state space, action space, reward function, and constraints are defined.
(10 Marks)
2. Implement an ϵ -greedy Multi-Armed Bandit algorithm in Python to maximize rewards under a limited budget of 500 trials. Compare the effect of different ϵ values (0.1, 0.3, and 0.5) on exploration, exploitation, and constraint satisfaction.
(10 Marks)
3. Design a Markov Decision Process (MDP) for an autonomous robot that must reach a destination while minimizing energy consumption. Define the states, actions, transition probabilities, rewards, and energy constraints.
(10 Marks)
4. Using Python and TensorFlow/Keras, implement a value iteration algorithm to determine the optimal policy for a constrained grid environment where certain states are restricted. Explain how Bellman equations are used to obtain the optimal policy.
(10 Marks)
5. A warehouse robot has a battery capacity of only 30 minutes. Design an RL framework that enables the robot to complete the maximum number of deliveries without exceeding the battery limit. Explain the constraints, policy, and reward design.
(10 Marks)
6. Implement a Reinforcement Learning agent that controls traffic signals at an intersection while ensuring that emergency vehicles receive immediate priority. Explain how exploration, exploitation, and policy learning are modified to satisfy the safety constraint.
(10 Marks)
7. Develop a Python program using Keras/TensorFlow to simulate a reinforcement learning agent for packet routing in a wireless network with limited bandwidth. Explain how bandwidth constraints influence the reward function and policy optimization.
(10 Marks)
8. Analyse an experimental comparison between random action selection and ϵ -greedy action selection in a Multi-Armed Bandit problem under a fixed time

constraint. Record cumulative rewards and explain which strategy provides better performance.

(10 Marks)

9. Design a Reinforcement Learning framework for a delivery drone that must maximize successful deliveries while avoiding no-fly zones and maintaining battery limits. Explain the MDP formulation, Bellman equation, and policy learning process.

(10 Marks)

10. Implement a Reinforcement Learning solution using Python for a smart energy management system where electricity consumption must remain below a specified threshold. Explain how reward shaping, policy learning, and feasibility constraints improve decision-making.

(10 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

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course: MLAI0303 - Reinforcement Learning

Constraint-Based

1. problem understanding: The objective is to design a RL agent that learns the shortest path from the start position to goal while avoiding obstacles in a grid world environment. The agent interacts with environment & improves its policy based on rewards.

constraint Analysis:

- The agent cannot move outside the grid boundaries
- obstacle cells cannot be entered.
- The episode ends when goal is reached or max no. of steps is exceeded.
- The agent should minimize the no. of moves.

Solution development:

State Space: Each grid cell represent a state.

Ex: (5x5 grid)

$$S = \{(0,0), (0,1), \dots, (4,4)\}$$

Action space: The agent has four actions:

→ up

→ down

- left
- right

Reward function

- Goal reached = +100
- obstacle = 20
- each movement = -1
- Invalid move = -10

Application of RL concepts.

- Q-learning learns the optimal policy.
- ϵ -greedy balances exploration & exploitation.
- Rewards guide the agent towards the goal.

Justification: This model satisfies all constraints by avoiding obstacles, minimizing unnecessary movements and maximizing cumulative reward.

problem understanding:

The objective is to maximize total reward by selecting the best slot machine bandit within only 500 trials.

Constraint Analysis:

- Total trials = 500
- Exploration should not waste too many trials.
- Exploration should maximize rewards.

Solution development:

The ϵ -greedy multi-armed bandit algorithm is used to maximize rewards within a fixed budget of 500 trials. During each trial, the algorithm either explores a random arm with probability ϵ or exploits the arm with the highest estimated reward.

Comparison				
ϵ value	exploration	exploitation	Reward	constraint satisfaction
0.1	low	high	Highest	Best
0.3	Medium	Medium	Good	Good
0.5	High	low	lower	Moderate

Application of concepts:

- small ϵ explores less & exploits more.
- large ϵ explores more but gains fewer rewards.

Justification: $\epsilon=0.1$ gives the highest cumulation rewards under the limited trial constraint because it quickly exploits the best arm.

3) problem understanding:

The robot must travel from start state to the destination while consuming the least amount of energy.

constraint Analysis:

- limited battery power.
- obstacles should be avoided.
- Minimum energy consumption.
- destination must be reached.

MDP components.

States Each location in the environment.

Ex: $S = \{ \text{start}, A, B, C, \text{Goal} \}$

Actions:

- Move up
- Move down
- Move left
- Move right.

Solution Development:

Transition probability:

Ex: $p(\text{next state} | \text{current state, action})$

correct move = 0.9

Slip = 0.1

Reward function

Goal = +100

Normal movements = -2

Obstacle = -30

Battery exhausted = -100

Energy constraint

$$\text{Energy} \leq 100 \text{ units}$$

→ Each movement consumes : 2 units

Application of concepts?

The robot chooses actions based on expected future rewards while considering energy limitations.

Justifications: The MDP optimizes long-term rewards while ensuring battery usage remains within limits.

problem understanding:

(*) The objective is to compute optimal policy in a constrained grid world using the value iteration algorithm.

Constraint Analysis:

Constraints:

- Restricted states cannot be entered.
- The shortest valid path should be selected.

→ Goal state must be reached.

Solution development:

The value iteration algorithm is applied to determine the optimal policy for a constrained grid world. Initially, all state values are assigned default values & are updated iteratively using Bellman Equation.

During each iteration, the value of every valid state is calculated by considering the maximum expected reward from all possible actions.

Bellman Equation:

$$V(s) = \max_a [R(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s')]]$$

Bellman Equation updates each state's value using:

→ Immediate reward

→ Discounted future reward.

The process repeats until values converge.

Application of concepts:

value iteration repeatedly applies Bellman updates to obtain optimal policy.

Justification: The algorithm guarantees an optimal policy while respecting restricted states & maximizing cumulative reward.

problem understanding:

The warehouse robot must maximize deliveries within a 30-min battery limit.

Constraint Analysis

constraints:

- Battery duration = 30 minutes
- Each movement consumes battery
- Robot must return to the charging station before battery depletion.

RL framework

State

- Robot position
- Battery level
- Delivery status

Actions

- Move forward, Turn left, right
- Deliver package
- Recharge

policy:

The agent selects actions that maximize deliveries while ensuring sufficient battery remains to reach the charging station.

Application of concepts.

The RL agent continuously learns the best delivery routes while considering battery constraints & maximizing cumulative rewards.

Justification: This framework satisfies the battery constraint, reduces unnecessary movement, increases successful deliveries & improves longterm operational efficiency.

